Numerical Simulations of AGN Accretion Disks in PLUTO

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Abstract

In this work, we present various two dimensional radiation hydrodynamic simulations of a thin viscous accretion disk around a nonrotating supermassive black hole with a mass of 10^8 $M_\odot$ in the PLUTO code. Starting from an existing star disk setup with viscosity and resistivity, we replace the central star with an absorbing inner boundary, use a Paczyński–Wiita instead of the original Newtonian potential, and add the nonrelativistic two moment radiation module with Kramers, Thomson and an effective Compton opacity. Once radiation is enabled, the advected passive tracer that restricted viscosity, resistivity and opacities to the disk no longer follows the disk material. We therefore develop a replacement called `DiskFraction` which classifies each cell from its density relative to adaptive radial reference profiles and based on its rotation. We validate this classifier against the tracer in short hydrodynamic, magnetohydrodynamic and resistive runs without radiation, and then use it in two long radiation runs at different resolutions. Both reach statistically steady states with accretion rates of about 0.07 to 0.09 times the Eddington rate and a radiative efficiency close to 0.048 in both cases. They carry quasiperiodic density waves that travel outward with a period of about 350 gravitational times, and show a weak inflow stream at intermediate altitudes. We discuss which of these features may be artifacts of the fixed sound speed in the viscosity prescription, of the boundaries and of the physics still missing from the setup.

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1 Introduction

Active Galactic Nuclei (AGN) are thought to be powered by accretion onto Super Massive Black Holes (SMBH) and make up some of the brightest objects in the universe. In the standard picture, the surrounding gas forms a geometrically thin disk in which an effective viscosity transports angular momentum outward and converts gravitational energy into heat that is radiated away locally (Shakura & Sunyaev 1973). For accretion rates around one tenth of the Eddington rate, the inner regions of such systems are dominated by radiation pressure, and in this regime the standard disk with viscous stress proportional to the pressure is thermally and viscously unstable (Lightman & Eardley 1974, Shakura & Sunyaev 1976). Whether real disks follow this prediction, and what happens in these cases, is a question that requires following the coupled evolution of gas and radiation in time, which is what numerical simulations enable us to do.

With this work, we build such a setup in the PLUTO code (Mignone et al. 2007). We start from an existing setup for the interaction of a magnetized star with its accretion disk (Čemeljić 2019) and adapt it to a black hole before enabling radiation as detailed in Section 2. Along the way, we find that the passive tracer used to separate disk and corona, and with it the physics restricted to the disk, stops working once radiation is enabled. Most of our effort therefore went into building a replacement for this tracer, the `DiskFraction` classifier, which we describe and validate in Section 3. Using this implementation, we then run the radiation setup at two resolutions until both reach a quasi steady state, and present first results in Section 4. The most important limitations of this setup are discussed in Section 5. Auxiliary equations, the analysis methods, supplementary figures as well as a link to all source files are included in the Appendix.

2 Numerical Setup

All simulations use version 4.4 of PLUTO (Mignone et al. 2007) on a two dimensional grid in spherical coordinates, with radius r and polar angle θ from the origin, as well as axisymmetry and equatorial symmetry, meaning that only one hemisphere is simulated. Additionally, we also use the cylindrical radius $R = r \sin\theta$ and height $z = r\cos\theta$ above the midplane throughout this text.

2.1 Base Model

The starting point is a stellar setup from a numerical atlas of such systems (Čemeljić 2019) which solves the equations of viscous and resistive magnetohydrodynamics for a thin disk around a rotating star with a magnetic field. Its initialization follows an analytic thin viscous disk (Kluźniak & Kita 2000) with aspect ratio $\epsilon = H/R$ and is embedded inside a corona in hydrostatic equilibrium. Viscosity and resistivity follow an α prescription and are restricted to the disk by a passive tracer, which is advected with the flow and initially equals one in the disk and zero in the corona. To ensure the disk stays thin without radiative cooling, the setup also includes an artificial cooling term, which S. Kundu and M. Čemeljić use in their setup. It removes a multiple of the local viscous and Ohmic heating from the gas, again weighted by the tracer, wherever this removal stays below ten percent of the thermal energy of the cell. Its exact form is given in Section A.

2.2 Black Hole Adaptation

Our initial tests ran on the stellar setups of the atlas. For a black hole, we replace the stellar surface at the inner radial boundary with an absorbing diode condition, which copies the state of the first active cell but lets gas and radiation only flow inward. The gravity of the central star is replaced with the pseudo Newtonian potential (Paczyński & Wiita 1980)

$$\Phi = -\frac{GM}{r - r_s} \tag{1}$$

where the term $r_s = 2GM/c^2$ gives the Schwarzschild radius. Within Newtonian dynamics, this potential reproduces the innermost stable circular orbit (ISCO) of a nonrotating black hole at six gravitational radii, and with it $\eta = 1/16$ for the efficiency of a disk that ends there. Lengths are measured in gravitational radii $r_g = GM/c^2$ and times in gravitational times $t_g = r_g/c$ where velocities have units of c accordingly. Without radiation, the equations solved here contain no other scale, so all results of the non radiative runs should hold for any black hole mass.

The domain extends from $r = 2.1$ to $r = 30$ in code units on a logarithmic radial grid and from the pole to the midplane on a uniform polar grid. The initial disk has $\epsilon = 0.1$ and an inner edge at $R = 6$ in these units, while the corona has the density profile $\rho_c \propto r^{-3/2}$ with a normalization of 0.01 in code units. At the outer boundary, the analytic disk is reinjected in a narrow wedge around the midplane, while inflow of coronal gas is blocked at higher latitudes. All tests of the tracer and the classifier in Section 3 were performed in this black hole setup, that is for a nonrotating black hole under Newtonian dynamics with the pseudo potential, with and without magnetic fields and resistivity.

2.3 Radiation & Physical Units

For radiation, we use the nonrelativistic version of the two moment radiation module of PLUTO with the full speed of light (Melon Fuksman & Mignone 2019, Melon Fuksman et al. 2021). Radiation introduces physical scales through the opacities as well as the radiation constant, so the units have to be fixed. We choose a black hole mass of $10^8\,\mathrm{M}_\odot$ for all radiation runs, which gives us units of $r_g \approx 1.48 \times 10^{13}$ cm and $t_g \approx 493$ s as the normalization. The density unit then is chosen to be $\rho_0 = 5.8 \times 10^{-10}\,\mathrm{g\,cm^{-3}}$ together with the viscosity parameter $\alpha = 0.5$ set in such a way that viscous heating and radiative cooling of the initial disk are roughly balanced, which corresponds to accretion at around a tenth of the Eddington rate. A twenty times higher density unit with $\alpha = 1$ gave a heating excess equivalent to strongly super Eddington accretion during a previous run, and the disk was disrupted within a few hundred gravitational times. In the radiation runs, the artificial cooling is removed, since heat is now carried away by the radiation.

Regarding opacities, we assume a standard hydrogen mass fraction $X = 0.7$ with a metallicity $Z = 0.02$ and include Kramers free free and bound free absorption (Kramers 1923) as well as a term $\kappa_{\mathrm{es}} = 0.2\,(1 + X)$ for Thomson scattering in $\mathrm{cm^2\,g^{-1}}$ units. PLUTO treats scattering as fully coherent, so in this form electron scattering never exchanges energy between gas and radiation. Inside AGN disks, however, it is Compton scattering that keeps the gas close to the radiation temperature, and without it viscous heating drove the gas in the inner disk to between 10^{10} and 10^{11} K in our tests, until overpressured bubbles broke the disk apart. We therefore add an effective absorption opacity

$$\kappa_C = \kappa_{\mathrm{es}} \frac{4 k_B T_r^4}{m_e c^2 \left(T_g + T_r\right)\left(T_g^2 + T_r^2\right)} \tag{2}$$

which has gas temperature T_g and radiation temperature T_r as inputs. Given this opacity, including the absorption term $\rho \kappa_C c \left(a_R T_g^4 - E_r\right)$ reproduces exactly the nonrelativistic Compton exchange rate

$$\Lambda_C = 4 \rho \kappa_{\mathrm{es}} c E_r \frac{k_B \left(T_g - T_r\right)}{m_e c^2} \tag{3}$$

since the factor $T_g - T_r$ cancels analytically against $T_g^4 - T_r^4$ in the absorption form. This rate is the usual approximation for Compton cooling and heating in radiation hydrodynamics, and terms of this form are included for example in radiation magnetohydrodynamic simulations of accretion disks (Jiang et al. 2019). The Compton part is moved from scattering to absorption, so the total extinction stays that of Thomson scattering and only the energy exchange changes. Several safeguards are added to complete the opacity model, these are listed in Section A. They hold the Compton rate fixed above gas temperatures of 10^9 K and switch energy exchange off in the corona and in gas that moves faster than about nine tenths of the speed of light. A floor of 10^4 K for the gas temperature prevents surface layers from cooling far below the range in which the opacity model is valid.

3 Disk Classifier

In the base model, the passive tracer decides where viscosity, resistivity, artificial cooling and later the opacities act. Once radiation is enabled, this tracer stops following the disk material in our runs. It stays at its initial distribution while the disk itself evolves, as discussed further below. We did not isolate the cause of this in the code. Instead, we replace the tracer with a classifier that derives the disk membership of each cell from its current state alone.

3.1 Construction

`DiskFraction` assigns each cell a value f between zero for corona and one for disk from two criteria. The first compares the density ρ of the cell with a corona reference density $\rho_c(r)$ and a disk reference density $\rho_d(r)$ at its radius, which are stored in 90 logarithmic radial bins and interpolated linearly in logarithmic radius and density. With the relative position of the cell between these references

$$s = \frac{\log \rho - \log \rho_c}{\max\left(\log \rho_d - \log \rho_c, 0.5\right)} \tag{4}$$

and the logistic function $\sigma(x) = 1/\left(1 + e^{-x}\right)$ the classifier reads

$$f = \sigma\left(\frac{s - 1/6}{0.3}\right) \sigma\left(\frac{|v_\phi|\sqrt{R} - 0.5}{0.1}\right) \tag{5}$$

where the second factor compares the rotation velocity v_ϕ with the Newtonian Keplerian velocity $R^{-1/2}$ in code units, so that only dense and rotating gas counts as disk. The midpoint of the density criterion lies one sixth of the way from the corona to the disk reference in logarithmic density. At the disk reference density itself, the first factor is only about 0.94 and f therefore stays slightly below one inside the disk. At the start of a run, before any profile has been updated, the classifier f equals the initial tracer exactly, since both are evaluated from the same condition that sets up the disk itself.

By default, the references are built from the classification itself. After every time step, the corona reference of a bin becomes the mean density weighted with $1 - f$ and the disk reference the mean density weighted with f over all cells in the bin. A disk estimate only counts if its weighted volume exceeds one thousandth of the bin volume, and bins that never had one copy the nearest valid bin. Both profiles are smoothed in time by an exponential moving average, which gives each new estimate the weight

$$w = 1 - \exp\left(-\frac{\Delta t}{\tau}\right) \tag{6}$$

with the time Δt between two updates and the time constant $\tau = 6\, t_g$ as parameters. The state of the classifier is written with every output, so that restarts continue with exactly the same classification.

3.2 Versions & Guards

A first version of the classifier gave each update the fixed weight 0.05 instead. Updated after every step, the profiles then had a memory of about twenty steps, which depends on resolution and corresponds to approximately 0.12 and 0.17 gravitational times for the two configurations described below. In a run with this version at the higher resolution, the density criterion inverted after the disk had been disrupted. In bins at radii between about 2.4 and 3.4 the disk reference density remained at about a factor of ten below the corona reference until the end of the run. Since the references kept getting built from the classification, this inversion sustained itself, due to dense gas classified as corona raising the corona reference and thin gas classified as disk lowering the mean disk value. The rotation criterion was not able to compensate, because any gas plunging inside the ISCO keeps its specific angular momentum of about 3.7 and therefore rotates two to three times faster than the Newtonian Keplerian velocity there. The result was an annular region close to the black hole classified as disk up to about 45° above the midplane, while the dense plunging inner stream lost its viscosity and the coupling to radiation.

Prompted by this, the current version adds multiple guards to the classifier. The logarithmic span between references used must be at least 0.5 dex as the denominator of s already includes above, so an inverted span can never flip the density criterion. After every update, bins with a smaller span copy both references from the nearest bin with a sufficient span, searching outward first, or have their disk reference raised to the minimum span if no such bins exist anywhere. Finally, the smoothing uses the time constant of Equation (6) instead of a fixed weight, so that the memory of the profiles is the same at every resolution and update cadence. Slight artifacts at the inner boundary still appear in this version, but the guards let the classifier repair itself instead of locking an error in place.

As a further option, the references can also be built without reference to the classification. The disk profile then takes the maximum density of each bin and the corona reference the geometric mean density of the cells above $z/R = 0.5$ in that bin. This mode cannot feed back on itself, but it has not been tested to the same degree as the default and is not used for the results presented here.

3.3 Validation

In testing the classifier, we ran the black hole setup without radiation as hydrodynamic, magnetohydrodynamic and resistive magnetohydrodynamic simulations with 90 by 45 cells up to $t = 1000$ in gravitational times. Each case was run twice. In the tracer driven run, the tracer f_{tr} decides where viscosity, resistivity and the artificial cooling act. In the classifier driven run, `DiskFraction` takes this role, while the tracer is still advected passively and can be compared. Figure 1 shows both for the hydrodynamic case. The two runs start identically and still look alike at $t = 250$ but from about $t = 500$ onward, the surface layer and the corona of the classifier driven run are somewhat thinner. Within the classifier driven run, tracer and classifier mark the same disk body, while the tracer extends further into the lifted surface layer. This gas has left the disk in density and rotation, so the classifier counts it as corona, while the tracer only remembers where it came from.

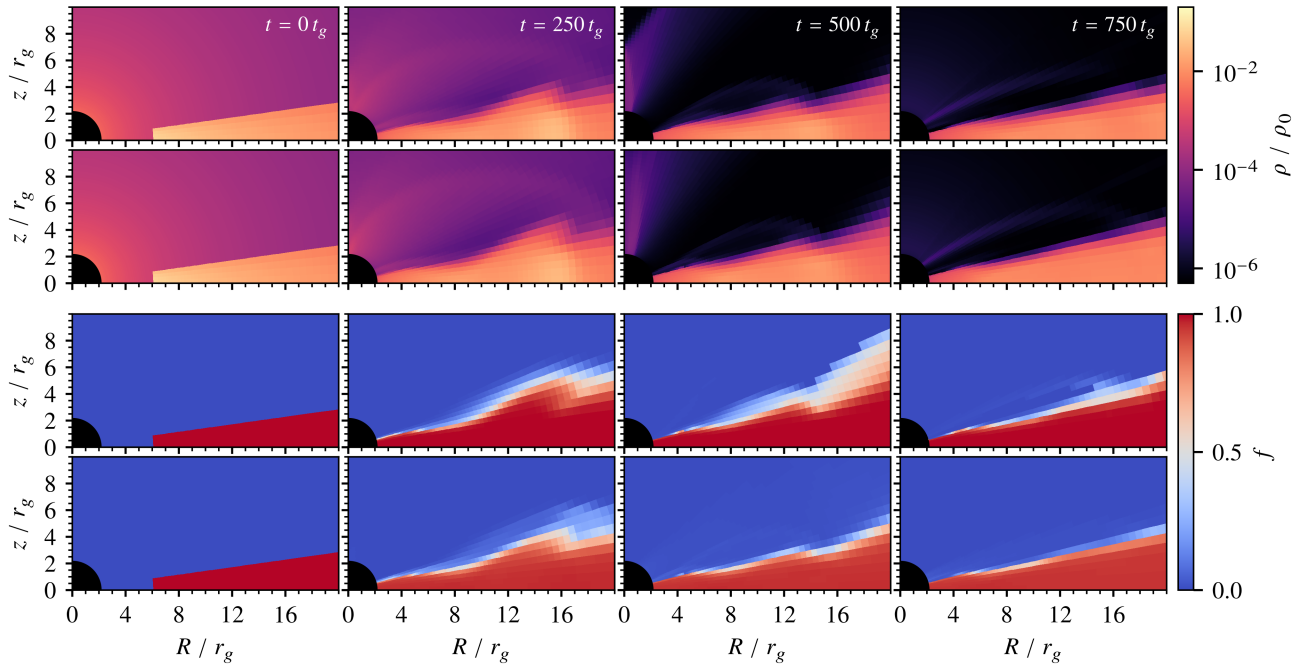


Figure 1: Hydrodynamic black hole setup without radiation. Density of the run driven by the passive tracer on top with that driven by `DiskFraction` below, and the passive tracer f_{tr} versus `DiskFraction` f_{df} within the latter run at the bottom.

Figure 2 quantifies this via the agreement metric $1 - \langle|\Delta f|\rangle$ with $\Delta f = f_{\mathrm{df}} - f_{\mathrm{tr}}$ averaged over the domain. Weighted by mass, the agreement stays around 95 % in all cases, and most of the remaining difference is due to f remaining slightly below one inside the disk region. Weighted by volume, the agreement dips to between about 84 % and 87 % around $t = 450$ while the surface layer is most disturbed, and in the hydrodynamic case falls to about 78 % at the end of the run. The maps of Δf in Figure 16 show negative differences almost everywhere, most concentrated near the surface layer.

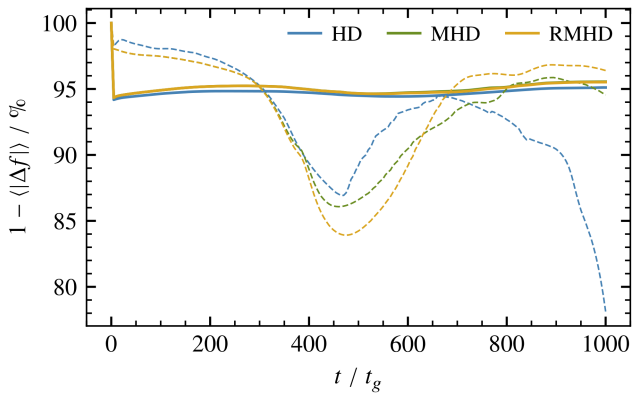


Figure 2: Match between f_{df} and f_{tr} in radiationless classifier driven runs, mass *(solid)* and volume *(dashed)* weighted.

With radiation, the picture changes completely. In the first radiation run, which still lacks any type of Compton coupling and in which `DiskFraction` already determines the physics, overpressured bubbles break up the disk, and the classifier marks the lifted, dense and rotating gas as disk material up to large heights. The passive tracer keeps its initial wedge throughout the entire run and no longer follows the gas at all, matching the previously described error mode.

4 Results

We now turn to the radiation runs with Compton coupling and the classifier included. The LR run uses 90 by 45 cells and reaches $t = 5 \times 10^4$ in code units, with outputs every 50 gravitational times. The HR run uses 120 by 90 cells and reaches $t = 4 \times 10^4$ with the same output cadence, which increases to 5 gravitational times from $t = 3 \times 10^4$ onward to enable a more accurate wave analysis. Apart from the resolution, the two runs differ in the version of the classifier, where LR still uses the first version described above, while HR uses the current version with all guards. Differences between the runs can therefore not be attributed to resolution alone. For the black hole mass of our setup, the run time of HR corresponds to about 230 days. The analysis methods are summarized in Section B.

4.1 Evolution & Relaxation

In the first few hundred gravitational times, the disk adjusts to the radiation. It starts to inflate, and a dense layer forms at its surface, which then breaks up into sheets that are lifted to several gravitational radii above the midplane.

After this, the two runs evolve differently, as Figure 4 shows. In the HR case, the disk is blown apart. Its mass falls from 132 to about 2 in units of $\rho_0 r_g^3$ by $t \approx 2000$ gravitational times, and the disk stays fragmented until about $t = 10500$ before it is rebuilt by about $t = 15000$ and settles at a mass of about 18 in the same units. For LR the disk mass instead declines smoothly from 143 to about 22 by $t \approx 5000$ without any breakup. We think this is most probably due to the stronger numerical viscosity, diffusivity and damping at lower resolution, which can smooth out the perturbations that grow into the HR breakup, although the different classifier versions may also contribute. In both runs, the corona loses most of its mass early on and is not replenished, since the outer boundary blocks coronal inflow.

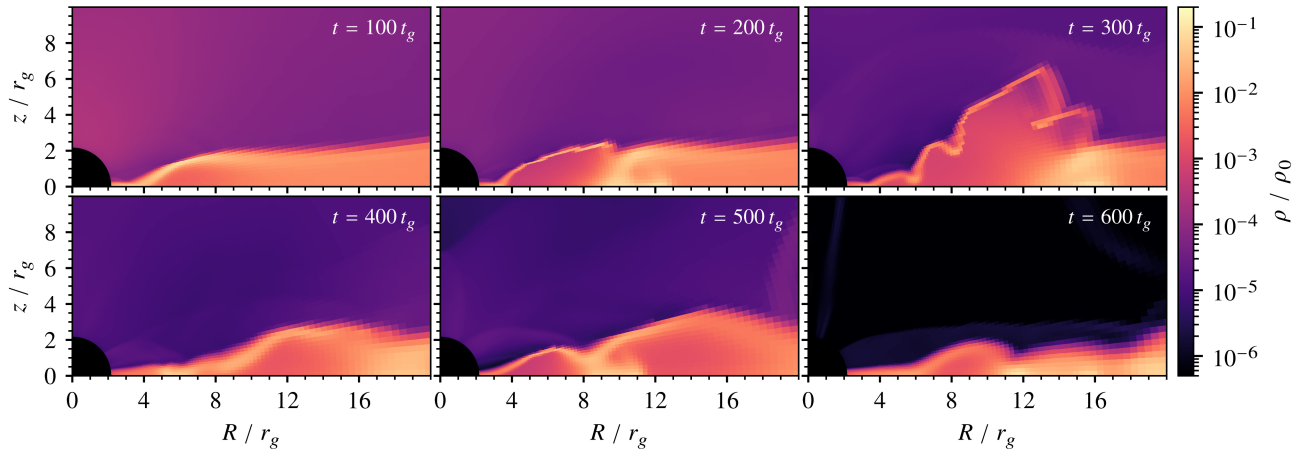


Figure 3: Density of HR in the early evolution. The disk inflates and forms a dense surface layer that breaks up.

To decide when a run has reached a steady state, we compare running means of the disk mass, the radiation energy, the accretion rate at the inner boundary and the luminosity at the outer boundary with their mean over the last fifth of the run. The steady state begins when all four stay within two standard errors of the reference, which happens at $t = 23725$ for HR and $t = 9225$ for LR in gravitational times. All averages in the following are taken over these windows, and within them the drift of the averages is consistent with zero.

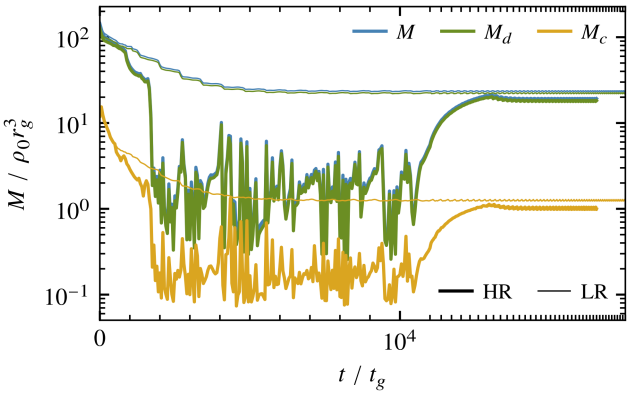


Figure 4: Mass split by `DiskFraction` for both radiation runs. Time is linear until $t = 1.1 \times 10^4$ and logarithmic after.

4.2 Steady State

Figure 6 depicts the initial density setup of HR next to its average over the last 10 000 gravitational times. In the steady state, the disk is thinner than initially. At $R = 10$ its aspect ratio is about 0.057 in HR and 0.042 in LR compared with 0.1 at the start, which means that LR resolves the disk with only about one cell per scale height against about three in the HR run. The corona has drained down to the density floor, and the unit optical depth surface lies inside the disk. The accretion rate in Figure 5 is flat in radius out to about $r = 17$ in both runs, so that the inner disk is in inflow equilibrium. Further out, the disk has not yet relaxed, which matches the estimate that viscous inflow reaches radii of 26 and 30 within the HR and LR run times. Beyond $r = 17$ we get a net accretion rate that is the difference of a mean inflow of roughly 0.2 Eddington rates and an outflow around 0.13 in the HR case, so the gas there circulates strongly. The corona carries twelve percent of the accretion at the inner boundary.

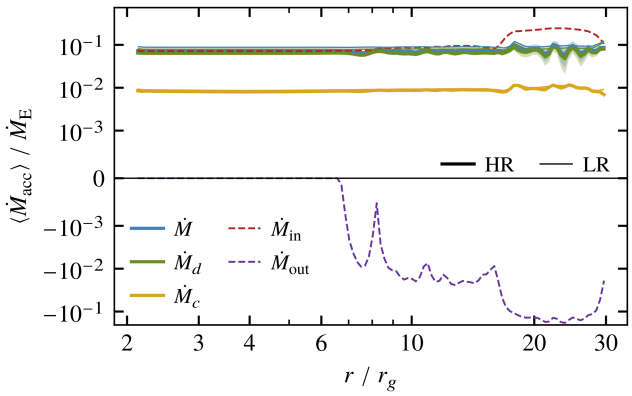


Figure 5: Time averaged accretion rate, positive for inflow, in total and split into disk versus corona, as well as inflows and outflows. Shaded bands show HR standard errors.

At the inner boundary, in Eddington normalized units, the mean accretion rate is 0.073 in HR and 0.091 in LR with the Eddington rate defined via the efficiency $\eta = 1/16$ from above. In solar masses, this corresponds to about 0.30 and 0.38 $\mathrm{M}_\odot\,\mathrm{yr}^{-1}$ respectively. The luminosity leaving the outer boundary is 0.056 and 0.070 times the Eddington luminosity, or about $8.3 \times 10^{44}\,\mathrm{erg\,s^{-1}}$ and $1.0 \times 10^{45}\,\mathrm{erg\,s^{-1}}$ between the runs. The ratio of both gives a radiative efficiency of about 0.048 in both runs, somewhat below the value for a disk that radiates all its binding energy. The vertical flux at the photosphere matches the vertically integrated viscous heating from about $r = 8$ outward, but falls short of it further in, as Figure 27 shows. Part of the heat released close to the black hole is therefore likely advected inward instead of being radiated, which is consistent with the lower efficiency.

For a steady thin disk with a stress free inner edge at six gravitational radii, the surface density follows

$$\Sigma_{\mathrm{ss}} = \frac{\dot{M}}{3\pi\nu}\left(1 - \sqrt{6/R}\right) \tag{7}$$

with the kinematic viscosity ν of our prescription defined in Section A. The measured surface density in Figure 7 lies below this profile at almost all radii except for a ring like maximum around $R = 17$ once the inflow equilibrium ends.

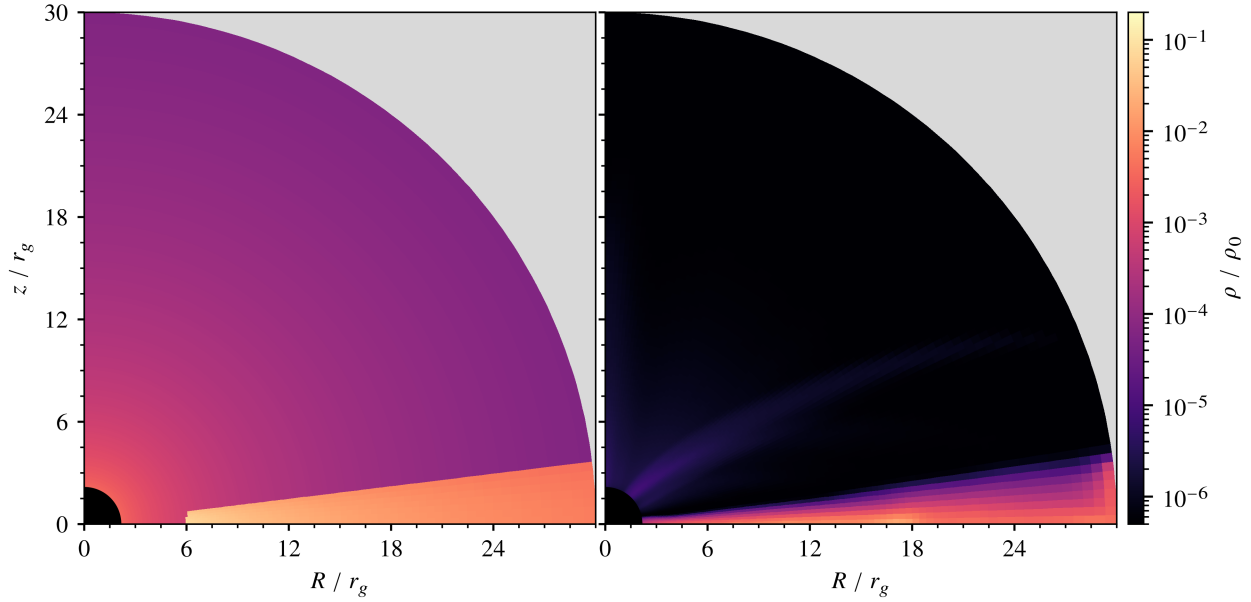


Figure 6: Density of HR run initially *(left)* and averaged *(right)* over the last 10 000 gravitational times.

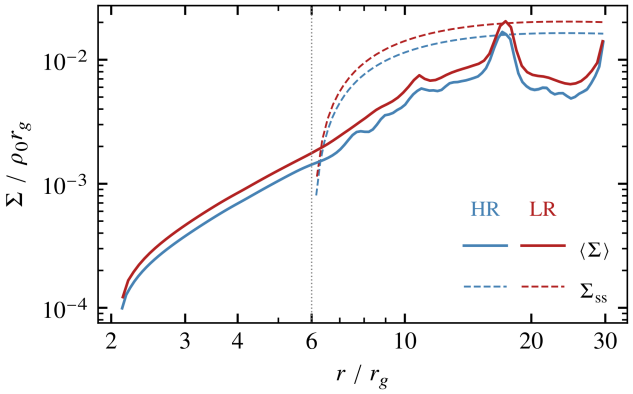


Figure 7: Time averaged surface density with steady state profile for given accretion rate. Shaded bands show errors.

A surface density below the steady state disk profile indicates that the transport of angular momentum is more efficient than the prescribed viscosity alone would allow. Inverting Equation (7) for an effective viscosity parameter gives values of about two to three times the prescribed α in both runs as shown in Figure 8 except for a narrow dip to about the prescribed value at the ring. The additional transport may come from resolved flows such as the waves below, or from numerical effects, which we cannot separate here.

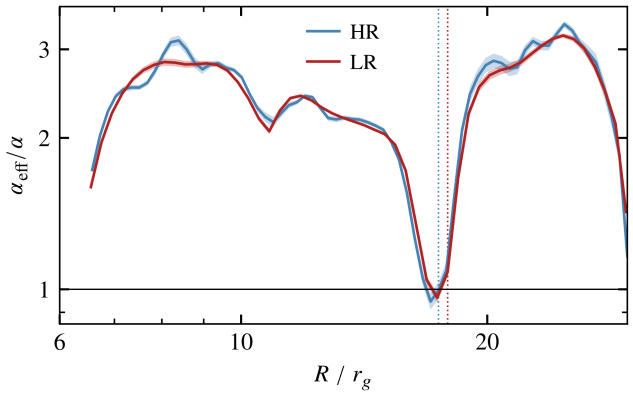


Figure 8: Effective viscosity parameter from the measured accretion rate and surface density, relative to the imposed value. Dotted lines mark inflow equilibrium radii per run.

The thermal balance of the disk is dominated by Compton exchange. Figure 9 compares the time over which viscous heating would double the thermal energy of the gas at the midplane with the corresponding cooling times by Compton exchange and by Kramers absorption. Between $R = 8$ and $R = 15$ the heating time sits at about 0.2 gravitational times, the Compton cooling time slightly shorter, and the Kramers cooling time two to three orders of magnitude longer.

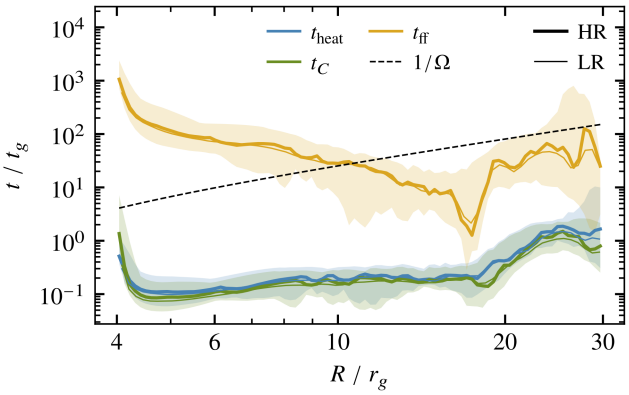


Figure 9: Midplane times for viscous heating, Compton and Kramers cooling, with the inverse orbital frequency for comparison. Lines are medians over the steady state window and bands show the standard error range for the HR run.

Heating and cooling therefore balance closely, with a ratio of about 0.8 to 0.9 as can be seen in Figure 10. This balance is reached far from local thermodynamic equilibrium, however. The median gas temperature at the midplane falls from about 1.7×10^8 K at $R = 6$ to roughly 10^7 K at $R = 16$ while the radiation temperature stays at about 8×10^4 K so that the gas is a hundred to almost two thousand times hotter than the radiation. Since the Compton rate grows with the difference of both temperatures, the gas has to heat up to these values before it can radiate the viscous heat, as Kramers absorption is far too weak in the present conditions.

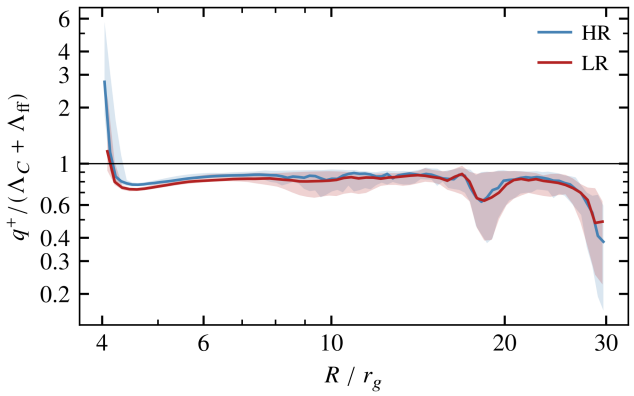


Figure 10: Ratio of viscous heating to radiative cooling at the midplane, with the standard error range as bands.

4.3 Density Waves

Both runs in their steady states exhibit quasiperiodic density waves that travel outward. Figure 11 shows relative surface density perturbations, where they appear as ridges that start near the ISCO and propagate outward to radii of $R = 17$ and beyond. Their root mean square amplitude, relative to the mean surface density, grows from about 0.25 near the inner boundary to about 0.8 around $R = 9$ and beyond $R = 20$ with a dip near $R = 15$ right inside the ring anomaly, making the waves strongly nonlinear. Power spectra of these perturbations peak at a frequency of roughly 2.9×10^{-3} per code time in both runs, equal to a period of 349 gravitational times or about two days for our SMBH mass.

The phase of the perturbation at this frequency grows steadily with radius as shown in Figure 12. Velocity and radiation energy perturbations at the midplane are roughly in phase with the density perturbation between radii of 6 and 12 as for an outward traveling compression, which we depict in Figure 25 in the Appendix.

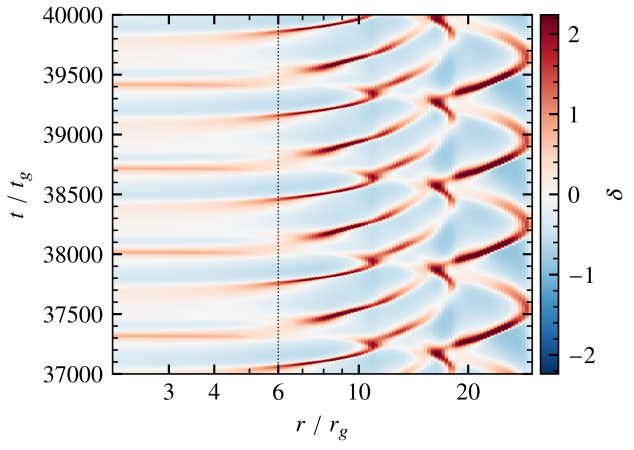


Figure 11: Relative surface density perturbation of HR over the last 3000 gravitational times at the dense output cadence. The dotted line marks the innermost stable orbit.

The phase speed in Figure 13 is about 0.019 times the speed of light between radii of about 4 and 17 in both runs. This is about 2.2 times the adiabatic sound speed including radiation pressure and 6.5 times the sound speed of the gas in HR alone. The waves are therefore not plain sound waves, and we have not identified their nature. That the period is unchanged in both runs, despite their different resolutions and classifier versions, suggests that it is set by the setup rather than by the numerics. It is close to the thermal time $1/(\alpha\Omega)$ with about 330 gravitational times at the outer boundary, which may hint at an origin there, possibly together with the fixed sound speed in the viscosity, but this remains to be tested. Beyond the ring at $R = 17$ the accretion rate and the enclosed mass oscillate with about 670 gravitational times instead, roughly twice the wave period.

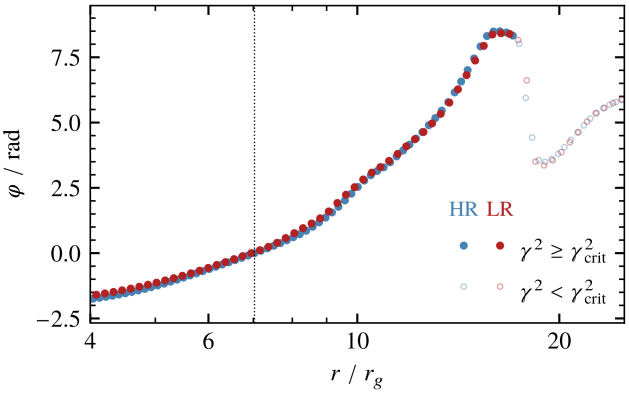


Figure 12: Phase of the surface density perturbation at the peak frequency relative to the dotted reference radius. Filled points have a significant coherence with the reference.

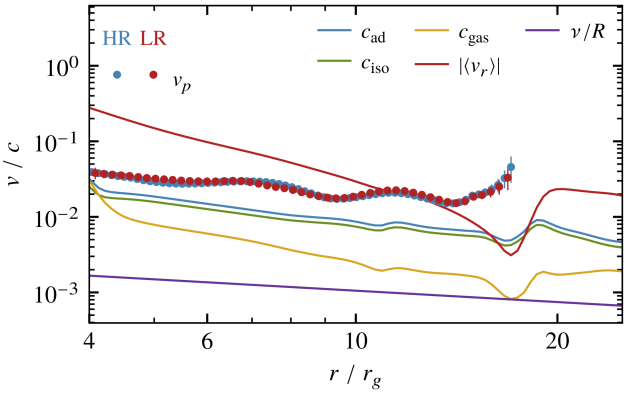


Figure 13: Local phase speed of the waves compared with the adiabatic, isothermal and gas sound speeds, the mean radial velocity and the viscous speed.

4.4 Inflow Stream

Most of the accretion happens close to the midplane. In the steady state of the HR run, roughly half of the net inflow passes within about one to three degrees of the midplane and nine tenths within about five to eight degrees at most radii, with $r = 10$ as an exception in HR at about twenty degrees, while the disk surface sits between five and nine degrees, so that about 90 % of the inflow passes through the disk. Above the disk, the angular distribution of the inflow in Figure 29 shows a second, much weaker maximum between about 20 and 50 degrees of elevation, at around a thousandth of the midplane value per degree. It belongs to a faint stream falling toward the central black hole at intermediate altitudes.

This feature is also visible as a diagonal shape in the averaged density of Figure 6 and may well be an artifact of the setup. The corona begins from Newtonian hydrostatic equilibrium but evolves in the deeper modified potential, and the outer boundary changes its treatment of flows into the domain at intermediate latitudes. We have not tested either explanation.

4.5 Resolution Comparison

Table 1 summarizes the main quantities of our radiative runs. The efficiency, the radius of inflow equilibrium and the wave properties agree within a few percent, while the accretion rate and luminosity of LR are somewhat higher by about a quarter, its disk is thinner, and it survives the early evolution without breaking up. Since the classifier version changes, we cannot definitively say which of these differences are due to resolution.

Table 1: Steady state quantities of both runs. Gravitational units are used for radii and times, rates are given in Eddington units, and velocities in terms of the speed of light.

Quantity	HR	LR
Cells ($r \times \theta$)	120×90	90×45
Classifier Version	current	first
Steady State Start	23 725	9225
Accretion Rate	0.073	0.091
Luminosity	0.056	0.070
Efficiency	0.048	0.049
Equilibrium Radius	17.4	17.9
Aspect Ratio ($R = 10$)	0.057	0.042
Cells / Scale Height	3.3	1.2
Wave Period	349	349
Phase Speed	0.0190	0.0195

5 Discussion and Outlook

Several features of the setup limit how far the results can be interpreted physically. Perhaps the most important one concerns the viscosity. Its sound speed is not derived from the evolving gas but instead from the initial disk profile at the same position, meaning it does not depend on pressure and is fixed in time. Stresses therefore cannot grow when the material heats up, which removes exactly the feedback responsible for driving the thermal and viscous instability of radiation pressure dominated disks (Lightman & Eardley 1974, Shakura & Sunyaev 1976). The stability of our steady states is thus no test of this instability.

The boundaries also affect the results. The disk reinjected at the outer boundary and the blocked coronal inflow set the supply of gas, the corona drains early and is not replenished, and the inflow stream discussed above as well as the wave period may originate there. The mass budget of the steady state does not close either. The change of enclosed mass and the integrated fluxes through the boundaries differ by about three percent between the radii 2.5 and 25 and by about 18 percent for the full domain, far beyond the sampling error and with a sign that the density floors, which can only add mass, cannot explain. This points to the boundary treatment or to the cell centered fluxes used in our diagnostic, as discussed in Section B.

Further limitations are due to suspicious or missing physics. The gas temperatures are probably too high for our setup to be fully trusted. Inside the disk, the matter stays two to three orders of magnitude hotter than the radiation, because only the Compton exchange can carry its heat away, and the corona reaches up to about 10^{11} K in our runs, where $k_B T$ is about 46 times the electron rest energy. The nonrelativistic Compton rate is far outside its validity there, which is why we hold it fixed above 10^9 K in the opacity model. The setup also includes neither relativistic corrections to the Compton scattering nor pair production, which would both matter at these temperatures. With Newtonian dynamics in the pseudo potential, the plunging gas inside about four gravitational radii moves superluminally, with radial velocities of up to about two times and rotation velocities of 1.4 times the speed of light at $r = 2.5$ in gravitational radii. The radiation there is decoupled from the gas to keep the nonrelativistic radiation module stable, but the dynamics of this region are unphysical. Finally, the simulations are axisymmetric and hydrodynamic, so turbulence driven by the magnetorotational instability is replaced by the α prescription, and LR barely resolves the disk vertically with only about one cell per scale height.

Natural next steps would be a viscosity which derives its sound speed from the evolving total pressure, and thereby allows the disk to become thermally unstable, a larger coarse domain that moves the outer boundary away from the region of interest, and an initial disk that is better suited to radiation pressure dominated accretion, such as a disk built from the standard thin disk solution (Shakura & Sunyaev 1973).

6 Conclusions

We adapted a viscous and resistive star disk setup in PLUTO to a nonrotating supermassive black hole and added radiation with effective Compton coupling. Since the passive tracer that restricts the disk physics fails once radiation is enabled, we developed the `DiskFraction` classifier, which agrees with the passive tracer to about 95 % by mass in runs without radiation, follows the disk material with radiation, and in its current version repairs its own errors. With it, two radiation runs at different resolutions reach steady states at roughly a tenth of the Eddington rate, in which Compton exchange balances the viscous heating closely, with a gas temperature far above the radiation temperature and the effective viscosity far exceeding the prescribed one by a factor of two to three, as well as strongly nonlinear density waves with a period of about 350 gravitational times traveling outward. Whether these features and the weak inflow stream at intermediate altitudes are physical or artifacts of the fixed sound speed, the boundaries and the missing high temperature physics has to be tested with the improvements outlined above.

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Appendix

This section contains auxiliary equations, the analysis methods, and supplementary graphics. All simulation setups, the analysis notebook and the plotting script for this report are found in this repository.

A Model Equations

All quantities in this section are in code units unless stated otherwise. The corona starts with the density $\rho = \rho_c r^{-3/2}$ and pressure $p = 0.4\rho_c r^{-5/2}$ where $\rho_c = 0.01$ is assumed. With the function

$$C = \frac{2}{5\epsilon^2}\left(\frac{1}{r} - \frac{1 - 5\epsilon^2/2}{R}\right) \tag{8}$$

the disk has the pressure $p = \epsilon^2 C^{5/2}$ and density $\rho = C^{3/2}$ wherever its pressure exceeds that of the corona and $R > 6$ holds, and its velocities follow the same analytic solution (Kluźniak & Kita 2000). In the radiation runs, the pressure of the analytic disk is shared between gas and radiation at a common temperature, which makes the initial disk radiation pressure dominated.

The dynamic shear viscosity is

$$\eta_\nu = \frac{2}{3}\alpha\rho\epsilon^2 C R^{3/2} f \tag{9}$$

where $\epsilon^2 C$ is the squared sound speed of the initial disk at the position of the cell, which is fixed in time. At the midplane, this produces the kinematic viscosity $\nu = 2/3\,\alpha\epsilon^2 R^{1/2}$ that Equation (7) uses. In the magnetic runs, the viscosity is also switched off where the ratio of thermal to poloidal magnetic pressure falls below 0.5 in a cell, and the resistivity follows an alpha prescription of the same form. In the tracer driven runs, the tracer takes the place of f here and in the cooling.

The artificial cooling of S. Kundu and M. Čemeljić removes in each step the energy

$$\Delta E = k\,\Delta t f\, q \tag{10}$$

from a cell, where q is the local viscous or Ohmic heating rate with $k = 2$ for viscous and $k = 1$ for Ohmic contributions. The removal is only applied where it is less than one tenth of the thermal energy density $p/(\gamma - 1)$ of the cell. It is not used in the radiation runs.

The Kramers opacity in cgs units is

$$\kappa_K = (K_{\rm ff} + K_{\rm bf})\,\rho\, T_g^{-7/2} \tag{11}$$

with $K_{\rm ff} = 3.68 \times 10^{22}\,(1 - Z)\,(1 + X)$ the free free and $K_{\rm bf} = 4.34 \times 10^{25} Z\,(1 + X)/10$ the bound free coefficients, where the factor ten in the second is a guillotine factor, and the gas temperature in this expression has at least 10^4 K in all cases. Above the $T_{\rm max} = 10^9$ K threshold, the Compton opacity of Equation (2) is replaced by

$$\kappa_C = \kappa_{\rm es}\frac{4k_B T_r^4\,(T_{\rm max} - T_r)}{m_e c^2\left(T_g^4 - T_r^4\right)} \tag{12}$$

which holds the exchange rate at its $T_g = T_{\rm max}$ value and matches Equation (2) there. It vanishes completely once the radiation temperature itself were to exceed the cap, while the code limits κ_C to $\kappa_{\rm es}$ in all cases so that the scattering opacity stays non negative. Since the rotation criterion never vanishes completely, `DiskFraction` leaves a value of about one thousandth in the corona. Together with the hot coronal gas, this was enough to Compton cool the corona in our tests, so the energy exchange uses the gated fraction

$$f_d = \max\left(\frac{f - 0.05}{0.95},\, 0\right) \tag{13}$$

instead. Near the black hole, the nonrelativistic radiation module becomes unstable for gas moving close to the speed of light, so all opacities are multiplied by the factor

$$g_\nu = \left[1 + \exp\left(\frac{\beta - 0.9}{0.03}\right)\right]^{-1} \tag{14}$$

with the gas speed $\beta = |\mathbf{v}|/c$ as input. The absorption and scattering opacities passed to PLUTO are then

$$\kappa_{\rm abs} = g_\nu f_d\,(\kappa_K + \kappa_C) \tag{15}$$

$$\kappa_{\rm sc} = g_\nu\,(\kappa_{\rm es} - f_d\kappa_C) \tag{16}$$

so that only Thomson scattering acts on all gas. In an earlier version, scattering was multiplying by `DiskFraction` also.

Radiation could then only push on gas classified as disk, and gas lifted across the disk surface lost its support and piled up into a dense sheet there, which motivated this change, although there was little change during short runs.

The density floor is set to $\rho = 5 \times 10^{-7}$ in code units, applied with a constant sound speed and momentum. At the inner boundary, all variables are copied from the first active cell, with radial velocity and radiation flux limited to inflow. At the outer boundary, density is extrapolated with a limited logarithmic slope that may not rise outward, the pressure follows adiabatically, and radiation leaves freely. Within an angle of $\arctan(1.25\epsilon)$ from the midplane the analytic disk is reinjected, and beyond $\arctan(3\epsilon)$ any inflowing velocity is set to zero.

B Analysis Methods

Time averages use the regular coarse output cadence of 50 gravitational times, and their standard errors are estimated from the scatter of block averages over 1000 gravitational times (Flyvbjerg & Petersen 1989), which accounts for the correlation between consecutive outputs.

For the relaxation of a global quantity X we compute running means $\bar{X}_W$ over windows of $W = 2500$ gravitational times, centered on t_c and compare them with the mean X_{ref} over the last fifth of the run through the statistic

$$D(t_c) = \frac{|\bar{X}_W(t_c) - X_{\mathrm{ref}}|}{\sigma} \tag{17}$$

where σ combines the standard errors of both means. Any quantity counts as relaxed from the first window after which D stays below two until the reference period, and the steady state begins when all four quantities in Section 4 are relaxed. Figure 32 shows this statistic for the HR run.

The radius of inflow equilibrium is the largest radius out to which the running mean of the accretion rate stays within twenty percent of its value inside six gravitational radii, and we quote here the median over the steady state window. The viscous estimate in Section 4 solves $t = R^2/\nu$ for the radius.

Power spectra as well as cross spectra of the surface density perturbation $\delta = \Sigma/\langle\Sigma\rangle - 1$ use the averaged periodogram method (Welch 1967) with Hann windows, half overlapping segments of at most 2500 gravitational timescales and linear detrending. The found peak frequency is refined by parabolic interpolation in logarithmic power. The phase of the cross spectrum with reference radius $r = 7$ at the peak frequency is unwrapped in radius, and only those radii with a coherence above its five percent significance level as well as a sufficient share of power at the peak are used. The local phase speed follows from a linear fit of the phase against radius over a range of fifteen percent in radius as $v_p = \omega/(\mathrm{d}\varphi/\mathrm{d}r)$ with the angular frequency ω of the peak.

The mass budget compares the change of mass between two radii with the time integral of the accretion rates through them. Both are taken from the dense outputs of HR where the accretion rates are evaluated at cell centers. Their difference would be the mass added by the density floor, and the error of the time integration is estimated from integrals over even and odd outputs separately. Figure 26 shows that this difference is negative, which the floor cannot cause.

The heating and cooling times of Figure 9 use the viscous heating rate term $q^+ = \eta_\nu \left(R\,\partial_R\Omega\right)^2$ at the midplane and the exchange terms of the gated Compton and Kramers opacities with the same gates as in the simulation, each relative to the thermal energy density of the gas.

C Runs

Table 2: Simulations used in this work. Runs marked with `TRC` are driven by `DiskFraction` and the others by the tracer.

Directory	Cells	End
`BH_VISC_HD`	90×45	1000
`BH_VISC_HD_TRC`	90×45	1000
`BH_VISC_MHD`	90×45	1000
`BH_VISC_MHD_TRC`	90×45	1000
`BH_VISC_RES_MHD`	90×45	1000
`BH_VISC_RES_MHD_TRC`	90×45	1000
`BH_VISC_RAD_HD_TRC`	90×45	1000
`BH_VISC_RAD_HD_TRC_COMP`	90×45	50 000
`BH_VISC_RAD_HD_TRC_COMP_FIX`	120×90	40 000

The last two are our LR and HR runs. In the first radiation run, a twenty times higher density unit is used, as well as a larger viscosity parameter $\alpha = 1$ and no Compton coupling. The runs without radiation use the same equations in code units, so their physical units do not matter.

D Supplementary Figures

The validation of the classifier for the magnetohydrodynamic and resistive cases is shown in Figures 14 and 15, and the difference between classifier and tracer for all three cases in Figure 16. Then follow global maps and early evolution panels of different quantities, as well as additional analysis metrics, such as accretion rates, amplitudes and phases of the waves, budgets and balances, vertical and angular profiles, relaxation statistics, and power spectra.

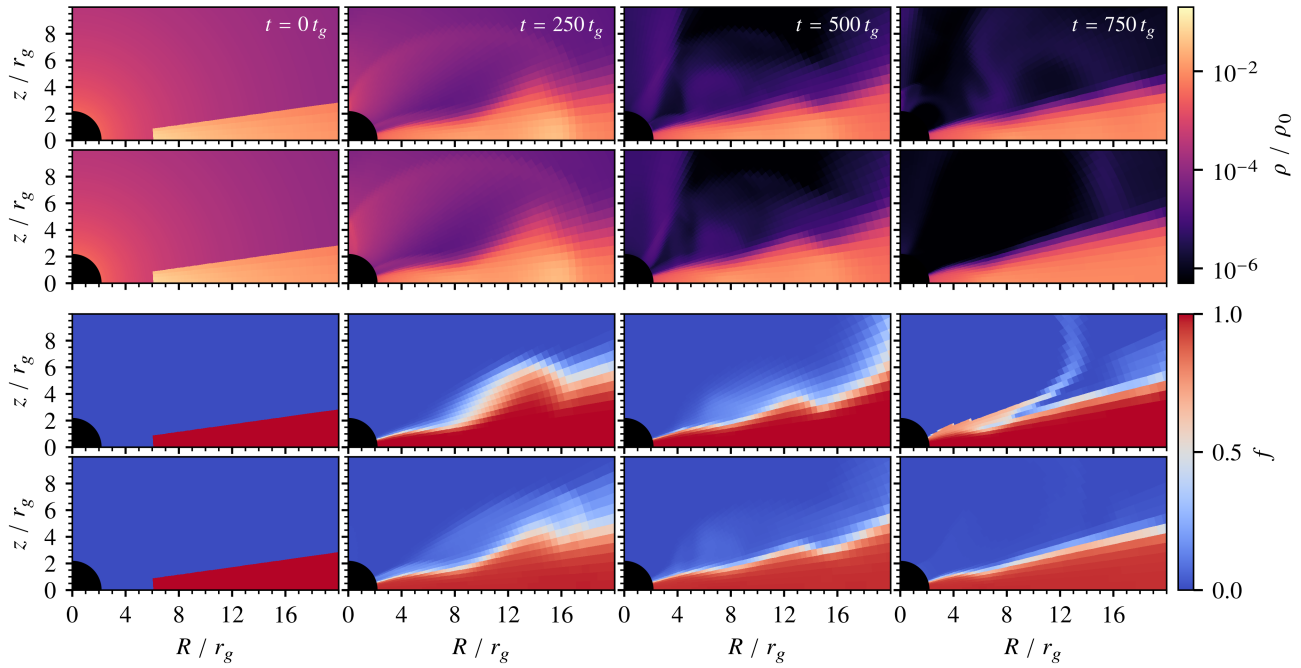


Figure 14: Magnetohydrodynamic black hole setup without radiation, top row driven by tracer, below by classifier, bottom tracer and classifier comparison.

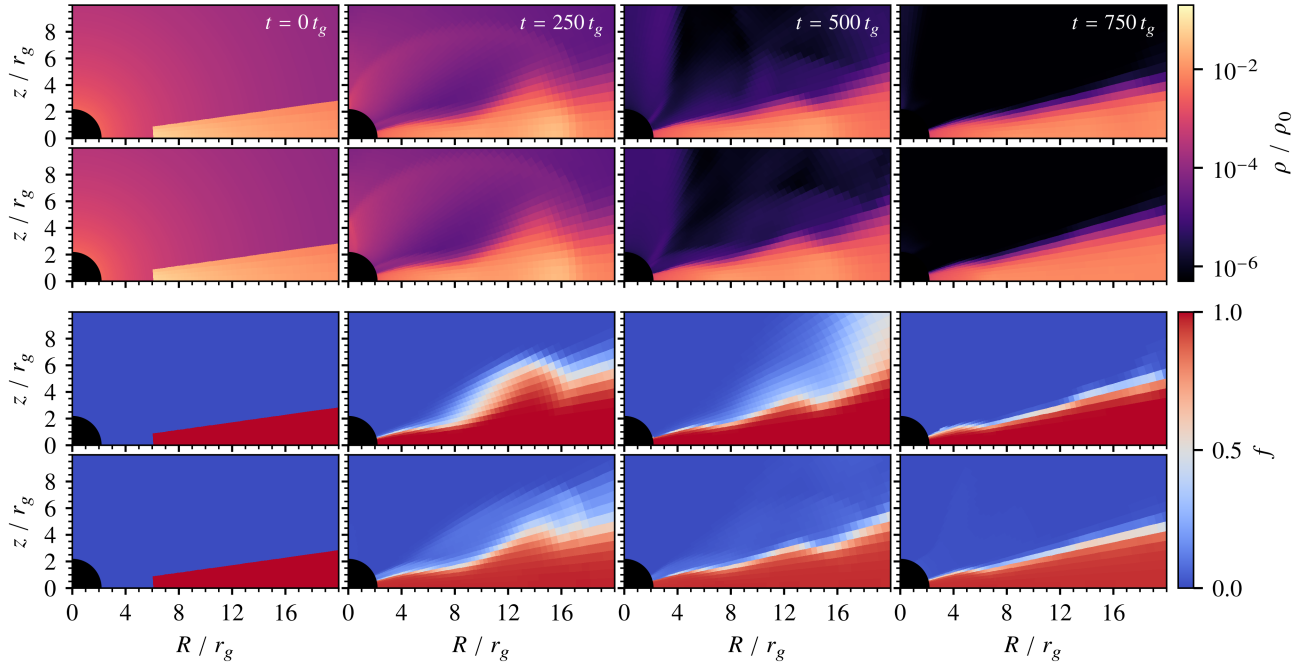


Figure 15: Resistive magnetohydrodynamic black hole setup without radiation, top row driven by tracer, below by classifier, bottom tracer and classifier comparison.

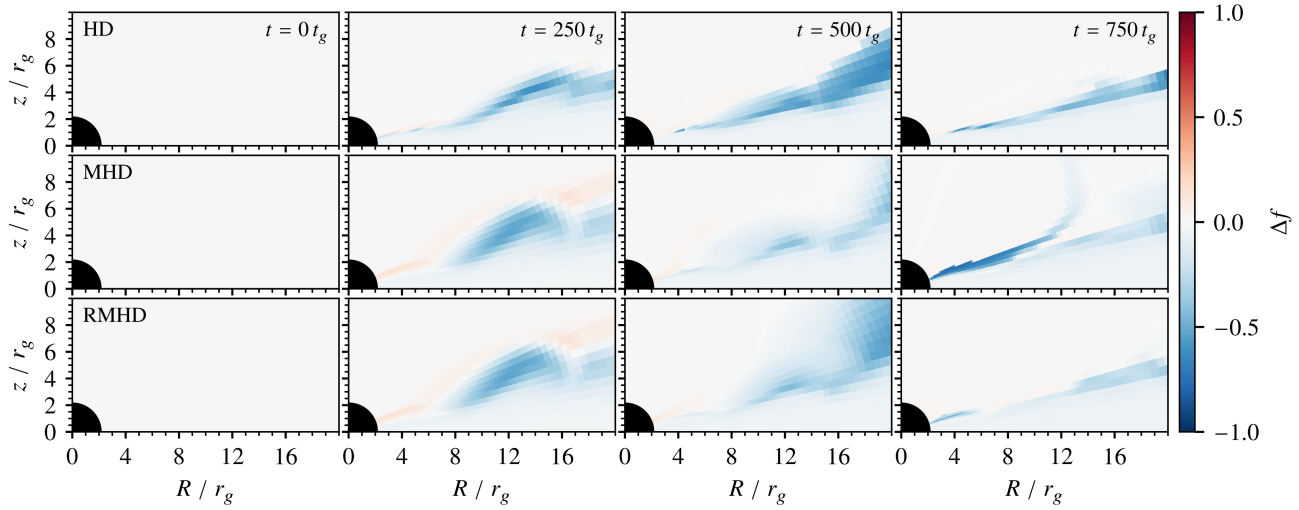


Figure 16: Difference $\Delta f = f_{\mathrm{df}} - f_{\mathrm{tr}}$ between `DiskFraction` and the advected tracer in the classifier driven hydrodynamic, magnetohydrodynamic and resistive runs.

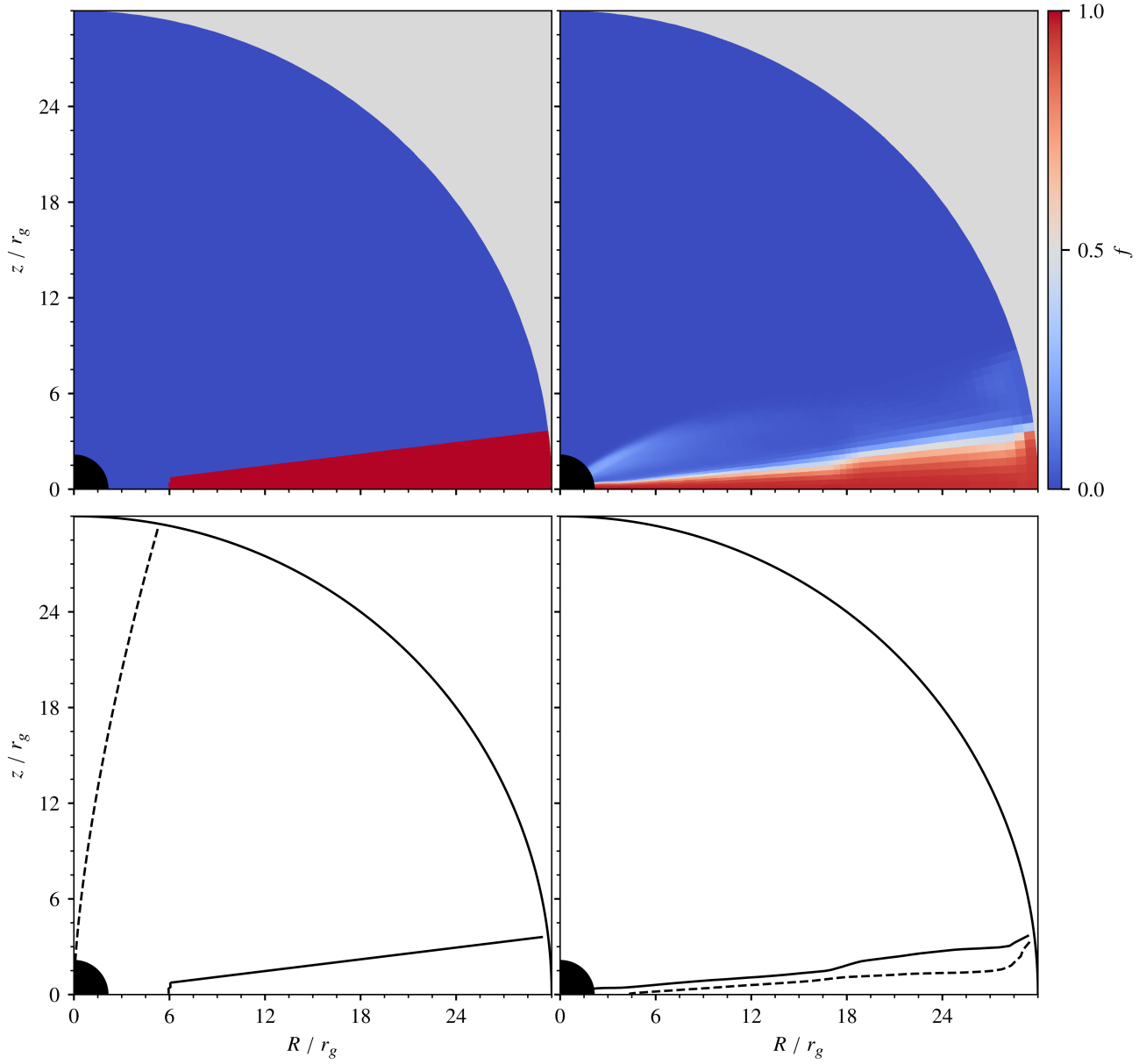


Figure 17: Classifier *(top)* as well as contours *(bottom)* of the disk surface where $f = 0.5$ holds *(solid)* and unit optical depth $\tau = 1$ from the pole *(dashed)* of HR initially *(left)* and averaged *(right)* over the last 10 000 gravitational times.

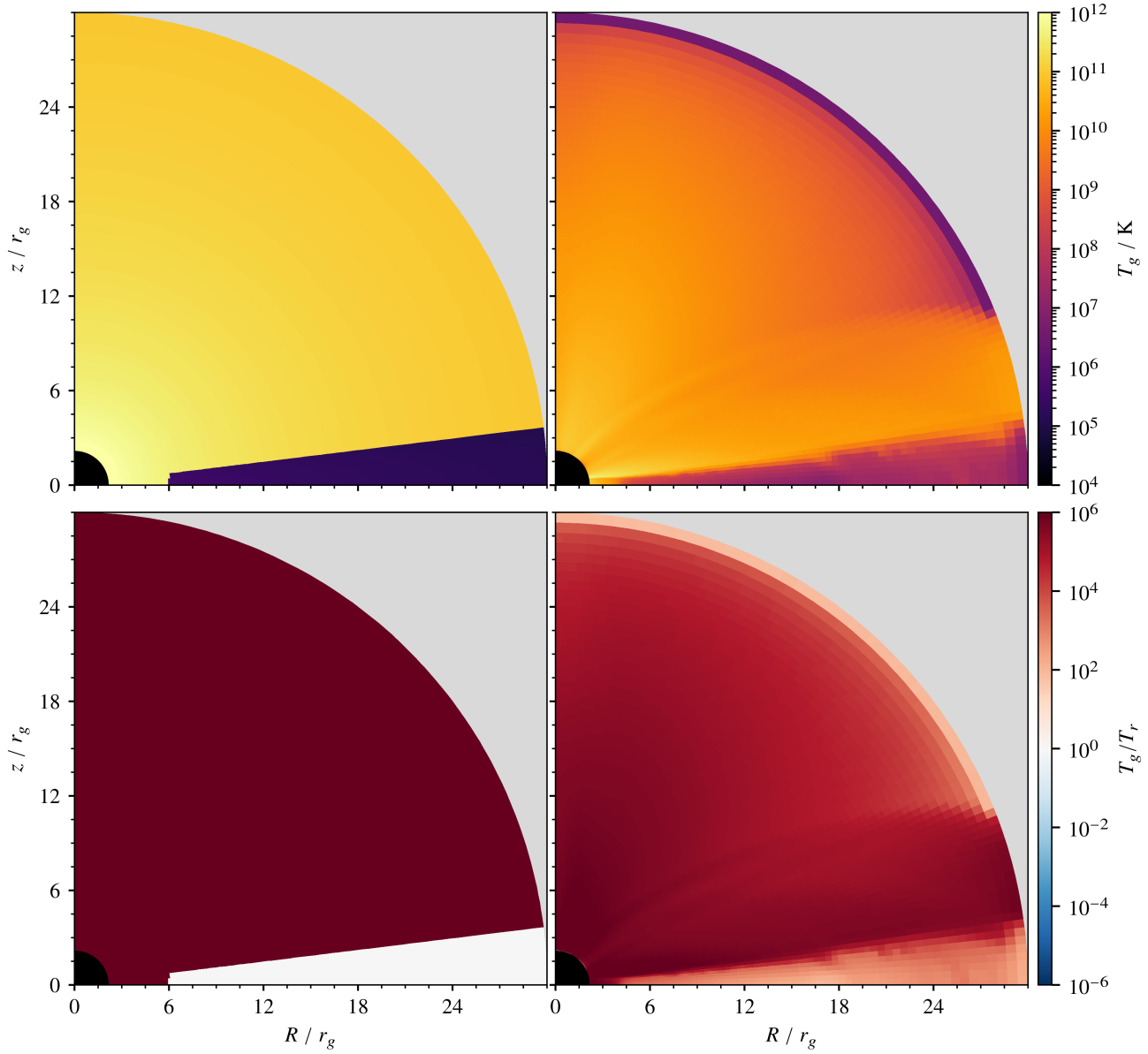


Figure 18: Gas temperature *(top)* and ratio of gas to radiation temperature *(bottom)* of the HR run initially *(left)* and averaged *(right)* over the last 10 000 gravitational times.

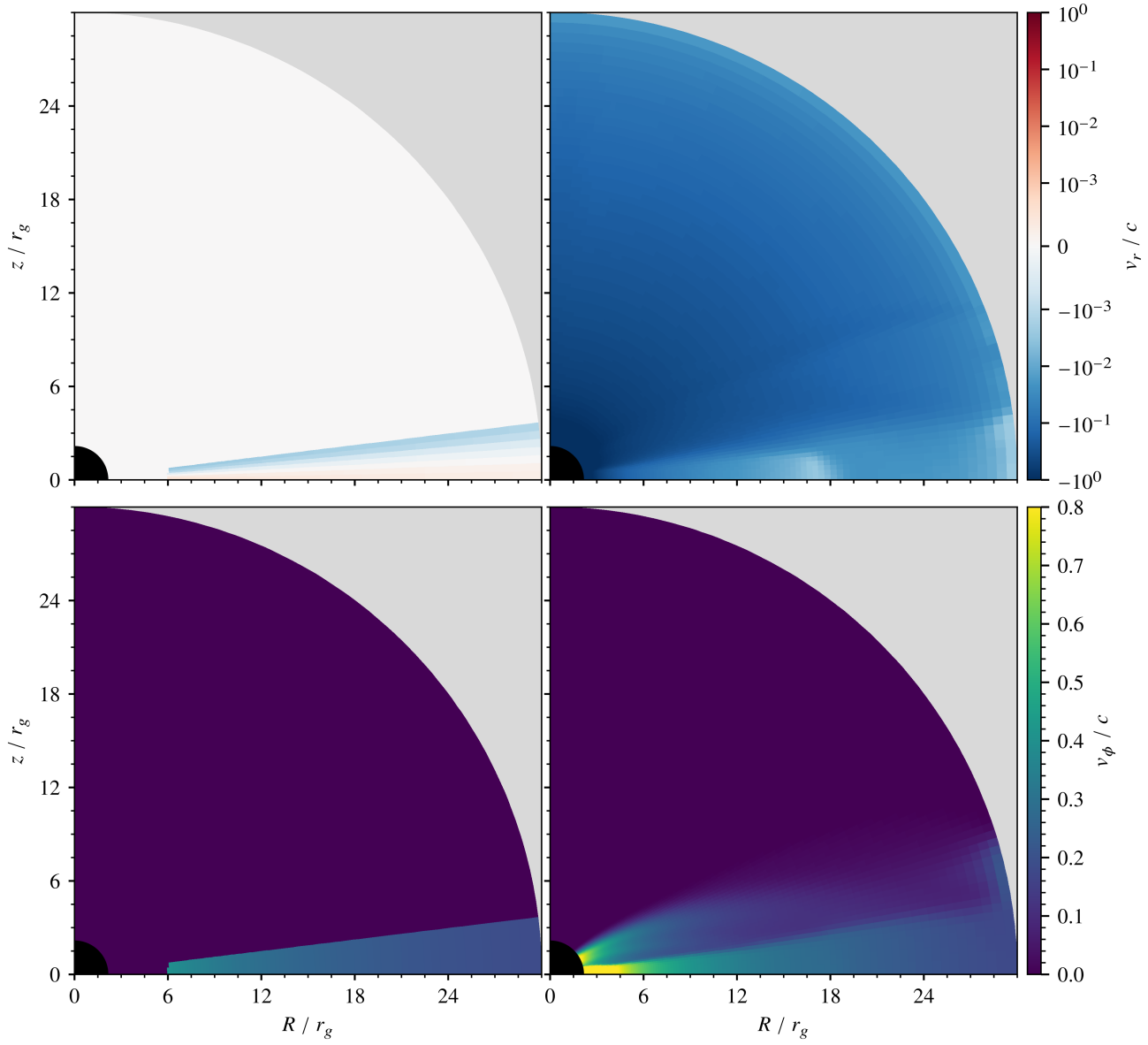


Figure 19: Radial *(top)* and rotation velocity *(bottom)* of the HR run initially *(left)* and averaged *(right)* over the last 10 000 gravitational times.

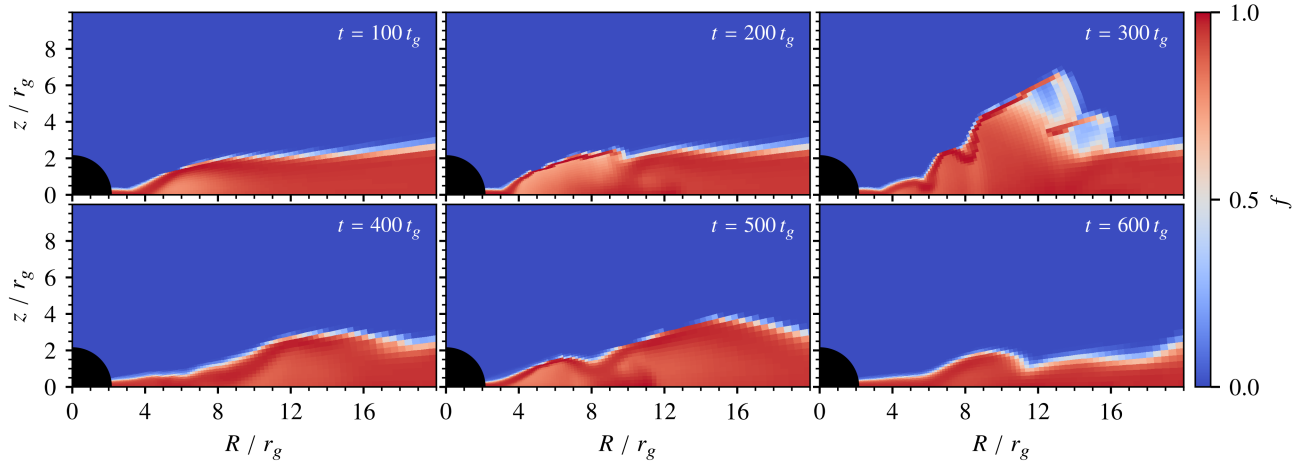


Figure 20: Evolution of `DiskFraction` in HR at early times.

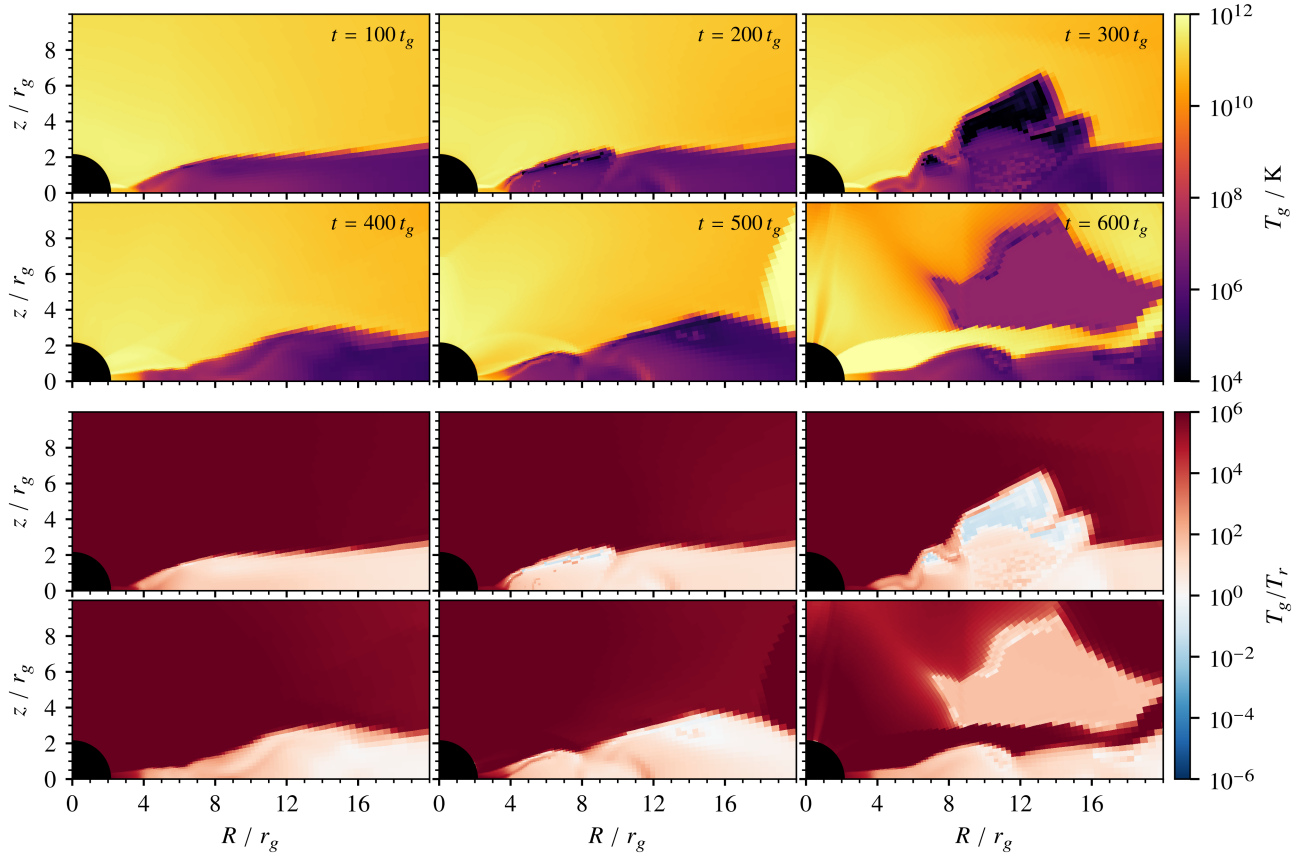


Figure 21: Gas temperature *(top)* and ratio of gas to radiation temperature *(bottom)* in the early HR evolution.

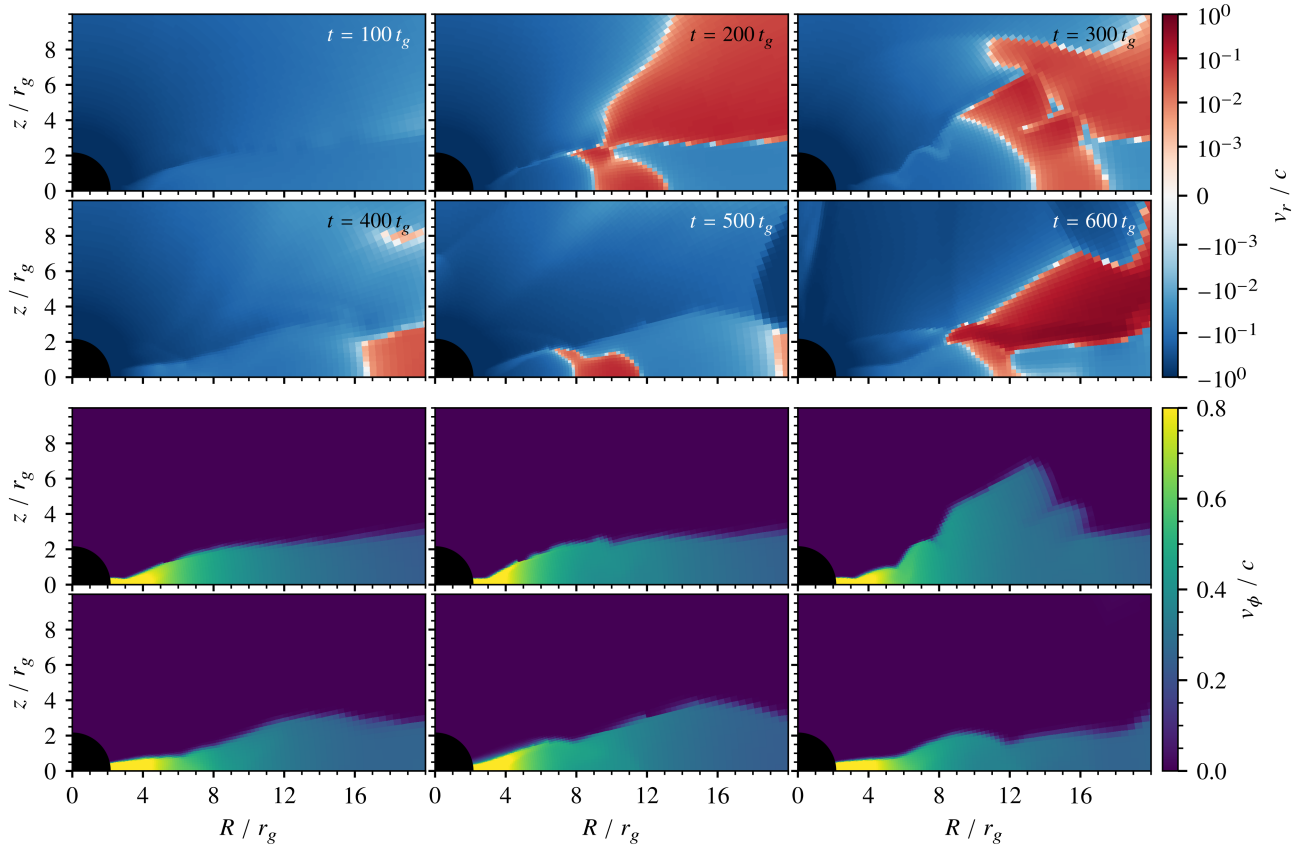


Figure 22: Radial *(top)* and rotation velocity *(bottom)* in the early HR evolution.

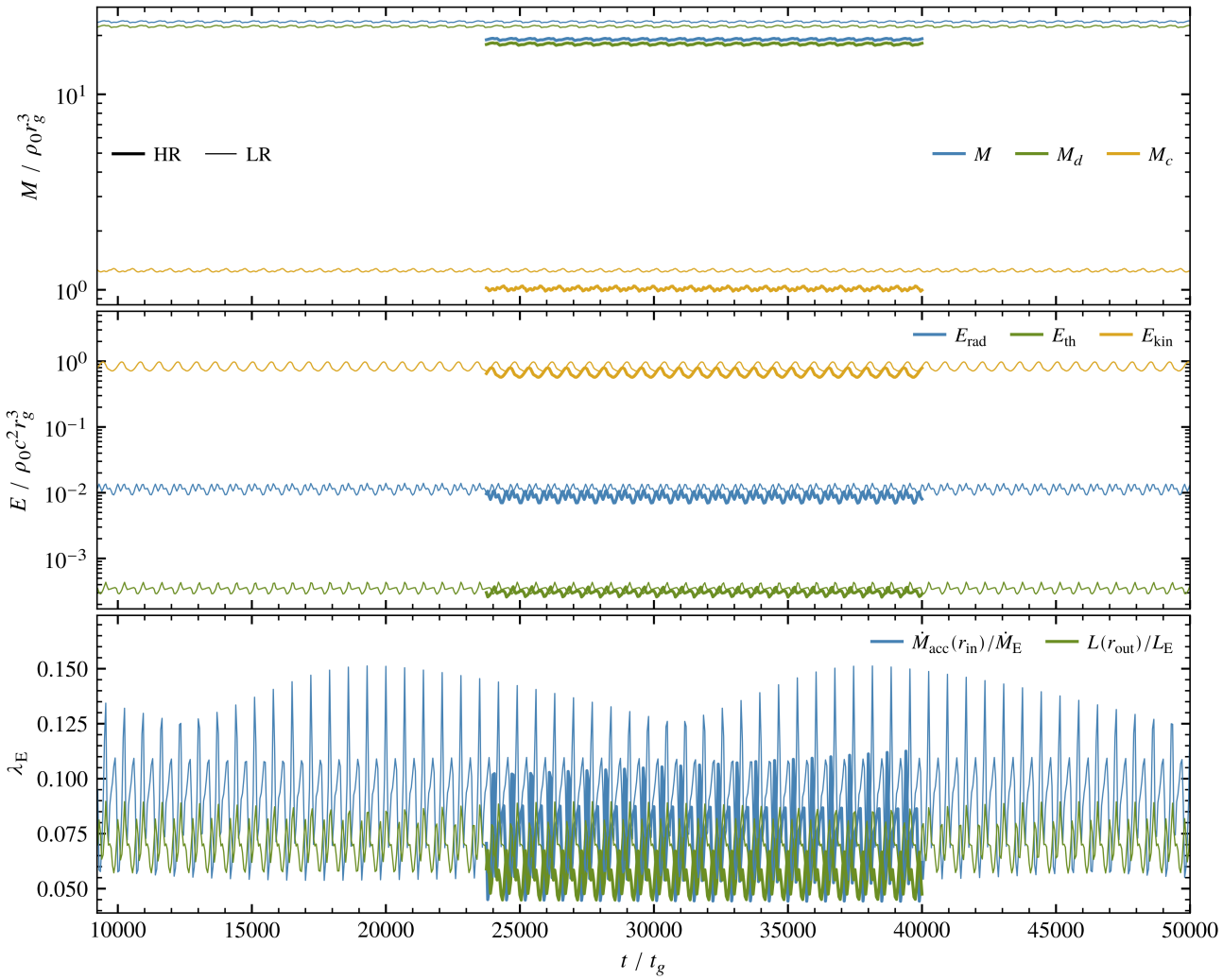


Figure 23: Masses, energies and Eddington ratios of accretion and luminosity of both runs in their steady states.

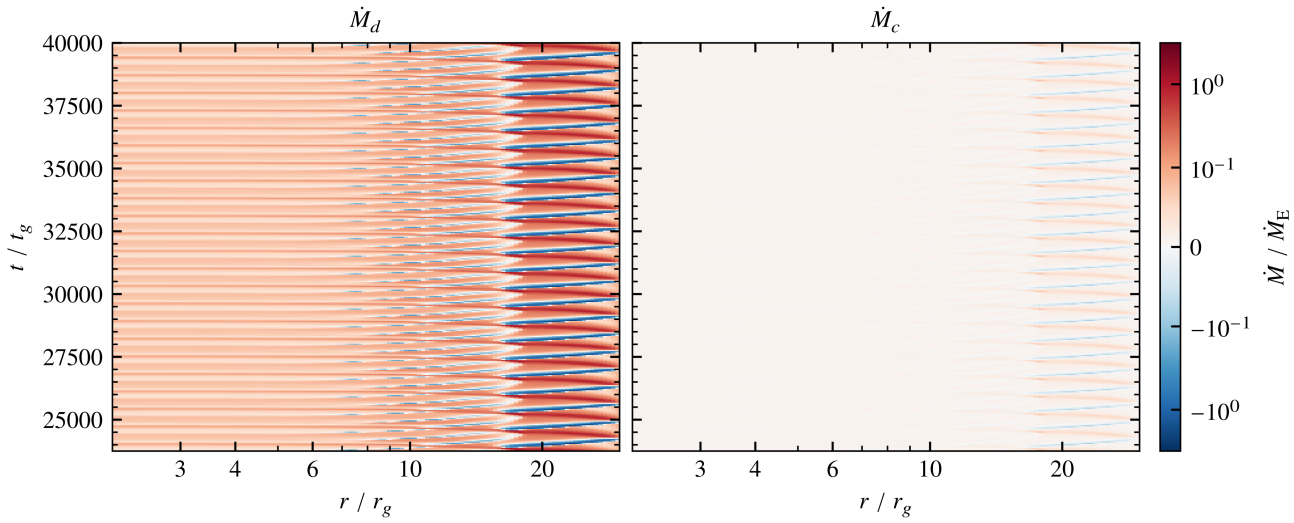


Figure 24: Accretion rate of disk *(left)* and corona *(right)* in HR over the steady state.

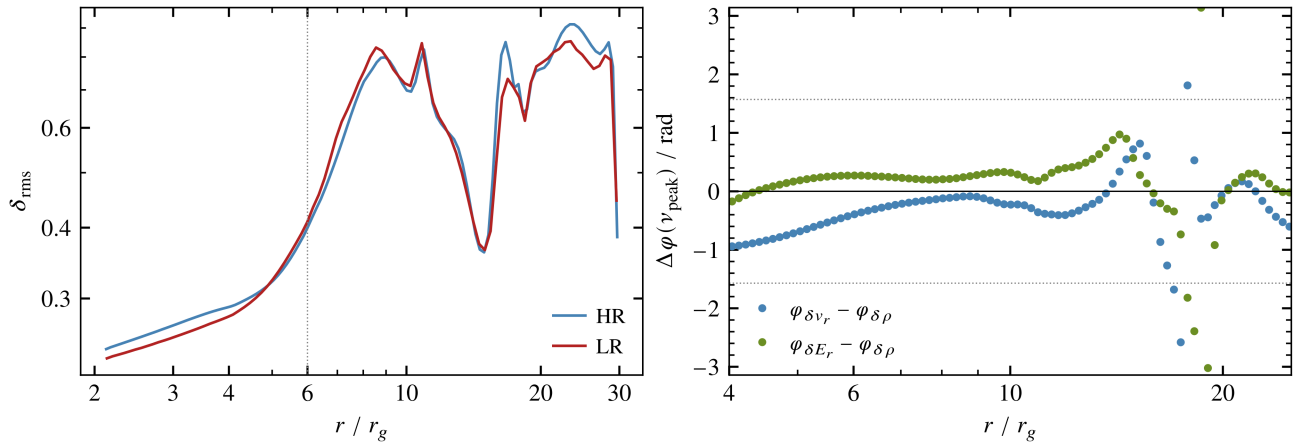


Figure 25: Root mean square of the surface density perturbation of both runs *(left)* next to phase differences of radial velocity and radiation energy relative to density at the midplane *(right)* in HR at the peak frequency.

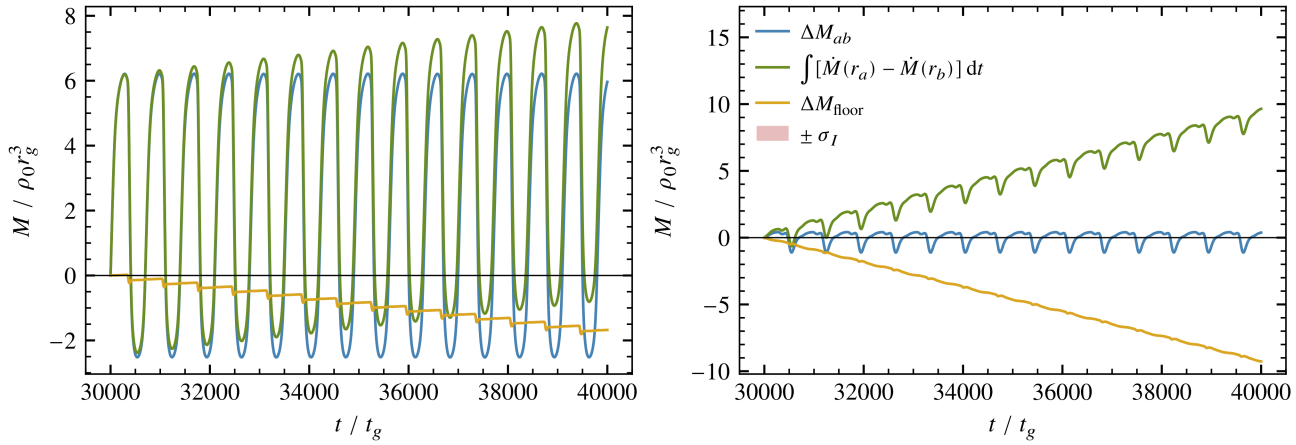


Figure 26: Mass budget of HR between radii 2.5 to 25 *(left)* and for the full *(right)* domain.

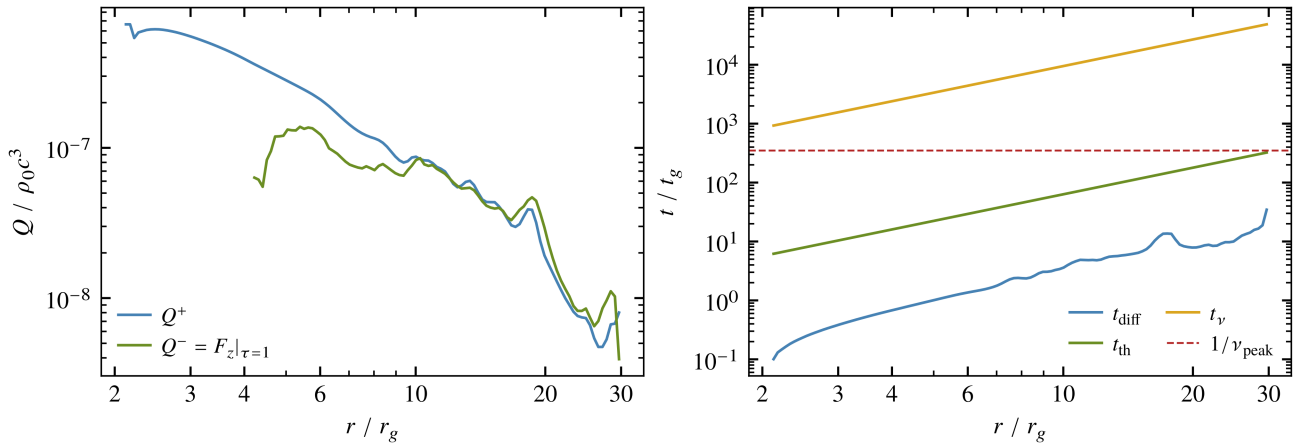


Figure 27: Vertically integrated viscous heating and vertical flux in HR at unit optical depth *(left)* as well as diffusion, thermal and viscous times *(right)* compared with the wave period.

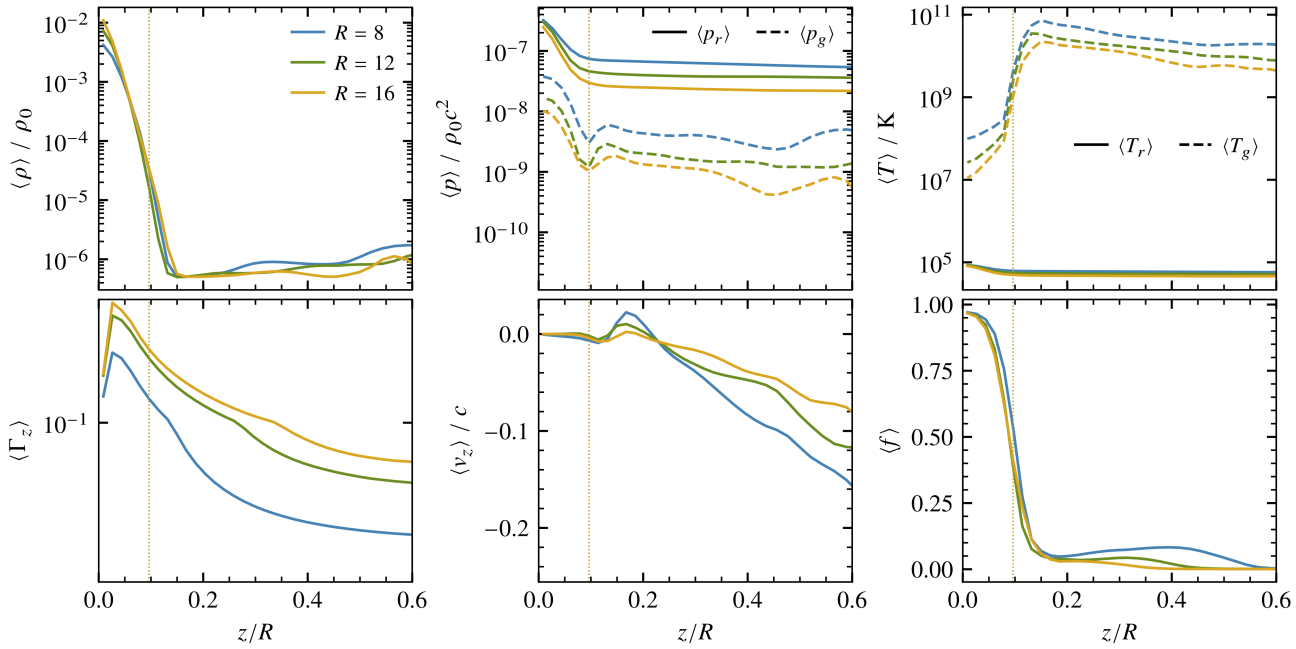


Figure 28: Time averaged vertical profiles of HR at different cylindrical radii, dotted lines mark the $f = 0.5$ disk surface.

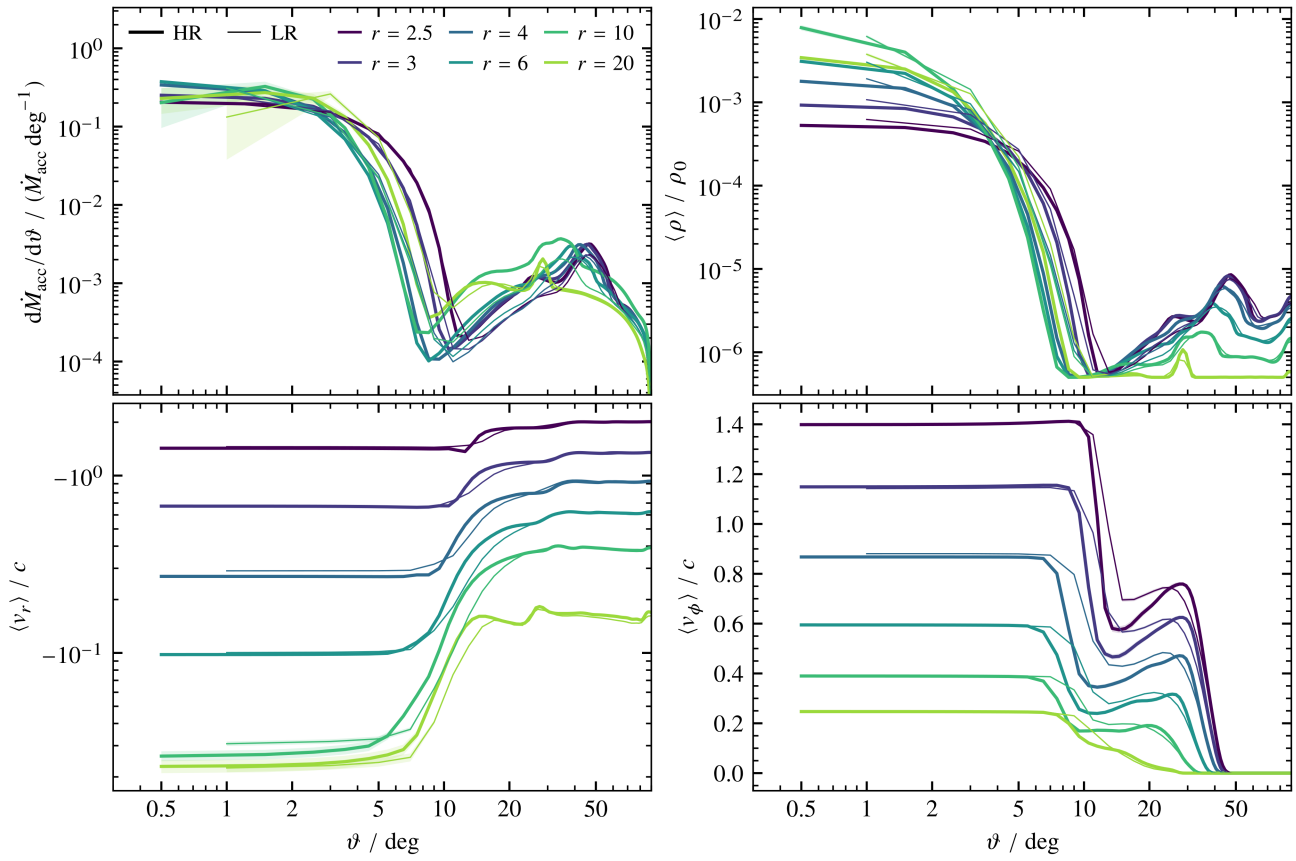


Figure 29: Time averaged angular profiles of net inflow, density, radial and rotation velocity at several radii.

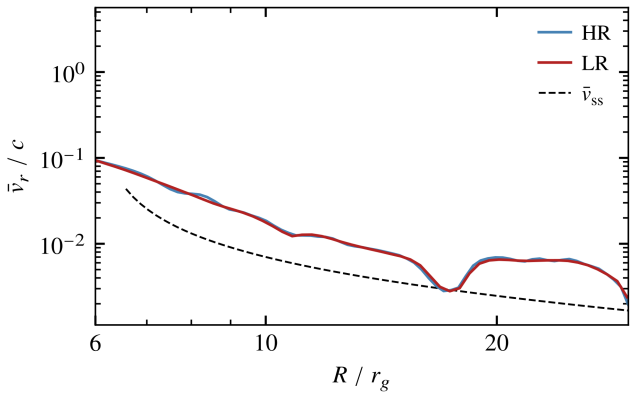


Figure 30: Mean radial inflow speed from accretion rate and surface density, compared with the steady state value of the prescribed viscosity used in our model.

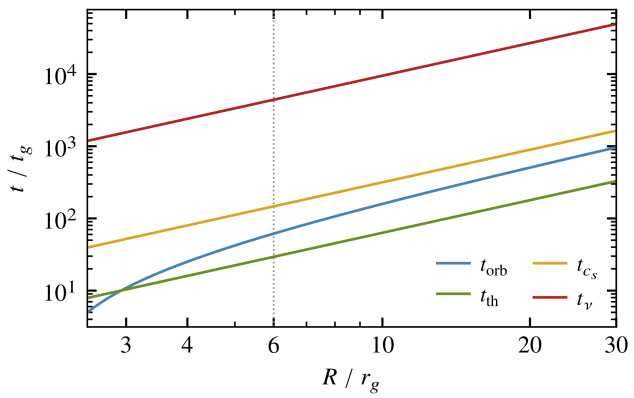


Figure 31: Orbital, thermal, sound crossing and viscous times of the disk seeding at initial time.

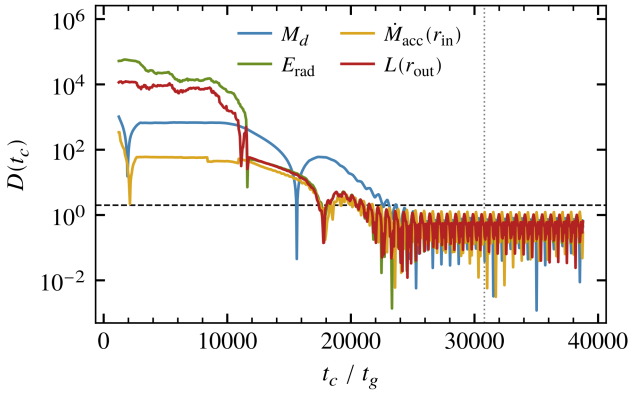


Figure 32: Relaxation statistic of global HR quantities, the dashed line marks the threshold and the dotted line the start of the reference period.

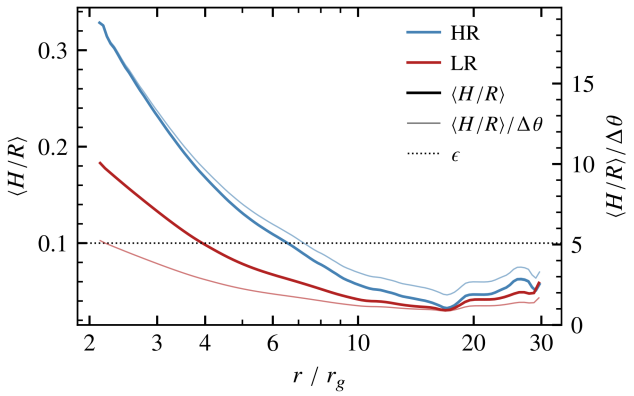


Figure 33: Time averaged aspect ratio of the disk and its resolution in cells per scale height.

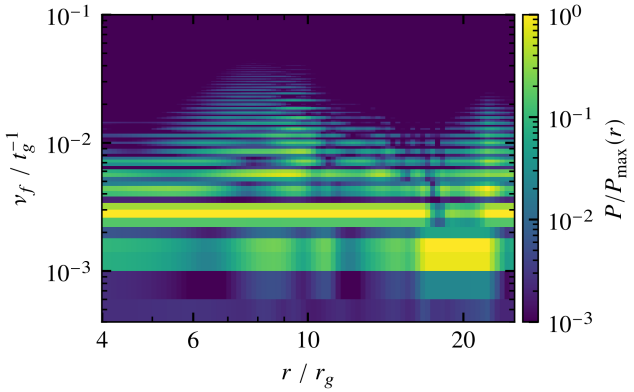


Figure 34: Power spectrum of surface density perturbation in HR at each radius, normalized to its maximum.